Basanta Khakurel

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Education

2026 Ph.D. in Earth and Environmental Sciences; Ludwig-Maximilians-Universität München

2023 M.S. in Biology; Southeastern Louisiana University (GPA: 4.0)

2021 B.S. (Honors) in Biology (Minors: Chemistry & Political Science); Southeastern Louisiana University

Research Experience

Sep 2023–Aug 2026: Höhna Lab

Affiliation: Department of Earth and Environmental Sciences, Ludwig-Maximilians-Universität München
Position: Research assistant; Ph.D. candidate
Ph.D. advisor: Sebastian Höhna
Dissertation title: *Toward more realistic models of morphological evolution in Bayesian phylogenetics*
Project: Bayesian phylogenetic methods for morphological datasets

Jun 2021–May 2023: Wright Lab

Affiliation: Department of Biological Sciences, Southeastern Louisiana University
Position: Graduate research assistant
M.S. advisor: April M. Wright
Thesis title: *Inference of phylogeny using discrete morphological characters*
Projects: Exploring morphological models for phylogenetic inference using `RevBayes` and `R`; posterior predictive simulations in phylogenetics

Aug 2019–May 2021: Wright Lab

Affiliation: Department of Biological Sciences, Southeastern Louisiana University
Position: Undergraduate research assistant
Advisor: April M. Wright
Honors thesis title: *Species delimitation of Eastern Pinesnakes (Pituophis melanoleucus)*
Project: Species delimitation analyses of the eastern Pinesnake complex

Publications

Google Scholar profile

Peer-reviewed publications

2026 Wynd BM, **Khakurel B**, Kammerer CF, Wagner PJ, Wright AM. Incorporating continuous characters in joint estimation of dicynodont phylogeny. *Systematic Biology*. doi:10.1093/sysbio/syaf078

2026 **Khakurel B**, Höhna S. A covarion model for phylogenetic estimation using discrete morphological datasets. *Systematic Biology*. doi:10.1093/sysbio/syag014

2026 Heckeberg N, Capobianco A, **Khakurel B**, Darlim G, Höhna S. Practical guide and review of fossil tip-dating in phylogenetics. *Systematic Biology*. doi:10.1093/sysbio/syaf050

2024 **Khakurel B**, Grigsby C, Tran TD, Zariwala J, Höhna S, Wright AM. The fundamental role of character coding in Bayesian morphological phylogenetics. *Systematic Biology*. doi:10.1093/sysbio/syae033

2024 Girón Duque JC, Balk M, Dahdul W, Lapp H, Mikó I, Alhajjar E, Wynd B, Tarasov S, Lawrence C, **Khakurel B**, Porto A, Yan L, Fluck IE, Porto D, Keating J, Borokini I, Seltmann K, Montanaro G, Mabee P. Meeting report for the Phenoscape TraitFest 2023 with comments on organising interdisciplinary meetings. *Biodiversity Information Science and Standards*. doi:10.3897/biss.8.115232

2023 **Khakurel B**, Nikolakis ZL, Crother BI, Wright AM. Species delimitation of eastern Pinesnake complex (*Pituophis melanoleucus*). *Bulletin of the Society of Systematic Biologists*. doi:10.18061/bssb.v2i1.9423

2022 Ward SJ, McMahan CD, **Khakurel B**, Wright AM, Piller KR. Genomic data support the taxonomic validity of Middle American livebearers *Poeciliopsis gracilis* and *Poeciliopsis pleurospilus* (Cyprinodontiformes: Poeciliidae). *PLoS ONE*. doi:10.1371/journal.pone.0262687

Manuscripts in review

Submitted **Khakurel B**, Capobianco A, Höhna S. The hidden cost of discretization: how the number of rate categories affects phylogenetic inference. Submitted to *Royal Society Open Science*.

Submitted **Khakurel B**, Darlim G, Capobianco A, Höhna S. Character level insights on the mosaic nature of morphological evolution. Submitted to *PNAS Nexus*.

Presentations & Posters

Invited presentations

2026 **Khakurel B**. Accounting for rate heterotachy in morphological phylogenetics: An overview in RevBayes. *Palaeoverse Lecture Series*, October, Online Webinar.

2026 **Khakurel B**, Höhna S. A covarion model for discrete morphological datasets. Beyond the Mk model: addressing heterotachy. *Bristol Palaeobiology Group*, University of Bristol, June, Online.

2024 **Khakurel B**. Snakes, fish, and Bayesian morphological phylogenetics. *PhD and PostDoc Seminar*, Paleontological Museum Munich, July, Munich, Germany.

2023 **Khakurel B**, Wynd BM, Wright AM. Use of continuous traits in phylogenetic estimation. *Departmental Seminar*, Department of Biological Sciences, Southeastern Louisiana University, April, Hammond, LA.

Contributed presentations

2026 **Khakurel B**, Capobianco A, Höhna S. The impact of rate variation assumptions on phylogenetic accuracy. *EvoGen Munich*, March, Munich, Germany.

2026 **Khakurel B**, Capobianco A, Höhna S. The hidden cost of discretization: is the default `k=4` biasing your tree lengths? *Society of Systematic Biologists Meeting*, January, Baton Rouge, LA.

2025 **Khakurel B**, Höhna S. Characters changing speed: the Covariomorph model and its impact on phylogenetic trees. *ESEB 2025 Congress*, August, Barcelona, Spain. (Poster)

2025 **Khakurel B**, Höhna S. Characters changing speed: the Covariomorph model and its impact on phylogenetic trees. *Evolution Meetings*, June, Athens, GA.

2024 **Khakurel B**, Höhna S. Evidence of covarion rate variation in empirical morphological datasets. *The Palaeontological Association Annual Meeting*, December, Erlangen, Germany. (Poster)

2024 **Khakurel B**, Höhna S. An efficient heterotachy model for morphological phylogenetics. *3rd Joint Congress on Evolutionary Biology*, July, Montréal, Canada.

2023 Heckeberg N, Capobianco A, Darlim G, **Khakurel B**, Höhna S. Potential and perils in using morphological characters for phylogenetic inference. *The Palaeontological Association Annual Meeting*, Cambridge, UK. (Poster)

2023 Wynd BM, **Khakurel B**, Wright AM. Incorporating continuous characters in total-evidence phylogenetic estimation of extinct lineages. *GSA Connects Meeting*, Pittsburgh, PA.

2023 **Khakurel B**, Wynd BM, Wright AM. Use of continuous traits in phylogenetic estimation. *Evolution Meetings*, June, Albuquerque, NM.

2023 **Khakurel B**, Wright AM. Model adequacy in morphological data. *Society of Systematic Biologists Meeting*, January, Ciudad de México, Mexico.

2022 **Khakurel B**, Wright AM. Species delimitation in eastern Pinesnakes (*Pituophis melanoleucus*). *Evolution Meetings*, June, Cleveland, OH.

2022 Schwery O, **Khakurel B**, Wright AM. Adequacy of fossilised birth-death models: a test case in ants. *Evolution Meetings*, June, Cleveland, OH.

2021 **Khakurel B**, Wright AM. Species delimitation in eastern Pinesnakes (*Pituophis melanoleucus*). *LBRN 8th Computational Biology and Bioinformatics Meeting*, Baton Rouge, LA.

2020 **Khakurel B**, Grigsby C, Tran T, Wright AM. Character state space partitioning in phylogenetic analyses from discrete data. *Louisiana Biomedical Research Network 18th Annual Meeting*, Baton Rouge, LA. (Poster)

2020 **Khakurel B**, Grigsby C, Wright AM. Modeling morphological state space. *Society of Systematic Biologists Meeting*, January, Gainesville, FL. (Poster)

2019 **Khakurel B**, Rocquin V, Litolff J. The role of honors student organizations in building and sustaining a robust honors presence on campus. *Southern Regional Honors Council Conference*, Memphis, TN.

Fellowships, Awards & Honors

2022 Graduate Research Fellowship. (Louisiana Biomedical Research Network). USD 8,000

2022 Distinguished Graduate Fellowship. (Southeastern Louisiana University)

2021 Outstanding Senior Award. (Southeastern Louisiana University)

2021 Dr. Marvin L. Yates Award of Excellence. (Southeastern Louisiana University)

2021 David Ware Outstanding International Student Award. (Southeastern Louisiana University)

2019 Sophomore Honors Distinction. (Southeastern Louisiana University)

Travel awards

2023 SSB Travel Award (Society of Systematic Biologists)

Honor societies

Member, Phi Kappa Phi Honor Society. (Southeastern Louisiana University chapter)

Member, Thirteen Club. (Southeastern Louisiana University)

Teaching

Jan 2026 Phylogenetic Analysis of Molecules and Morphology: *Modeling morphological evolution*. Invited lecturer. (Ludwig-Maximilians-Universität München)

Sep 2023 Biological Data Analysis: *Simulations in biology*. Invited lecturer. (Southeastern Louisiana University)

Summer 2019 Upward Bound Math and Science Program: taught mathematics and biology to high school students. Summer camp teacher. (Southeastern Louisiana University)

Service

Publons profile

Professional

Reviewer for *Methods in Ecology and Evolution*.

Outreach

2024–Present Project contributor, Himalayan Pariwar e.V., Munich, Germany: empowering accessible education in Nepal.

2021 Planning committee, Louisiana Sci-Comm Summit, Baton Rouge, LA.

2018 Volunteer judge, Southeastern Lab School Science Fair, Hammond, LA.

University

2018–2022 President, Nepalese Student Association at Southeastern.

2020–2021 Vice President, Phi Kappa Phi Honor Society, Southeastern chapter.

2019–2021 Ambassador, Honors Program at Southeastern.

2019–2020 Leadership Ambassador, Division of Student Affairs at Southeastern.

2019 Senator, Student Government Association at Southeastern.

Professional Organizations

Society for the Study of Evolution (SSE)

Society of Systematic Biologists (SSB)

Workshop & Training Participation

A Primer for Phylogenetic Causal Inference workshop.

Generalized Bayesian Phylogenetics to Infer Simultaneous and Multifurcating Divergences.

Phylogenetic Methods for Historical Biogeography.

Visualizing Science workshop series.

Navigating Scientific Publishing – Early Career Focus.

Phenoscape TraitFest.

RevBayes workshop and hackathon.

PhylogatR (Phylogeographic data Aggregation and Repurposing) workshop.

Posterior predictive simulation with P2C2M workshop.

Skills

Computing Shell scripting

Computer programming (`R`, `C++`, `Rust`)

Version control (Git)

Document markup (LaTeX, Typst, HTML/CSS)

Text editing and regular expressions (vim, sed, awk)

	HPC environments (Slurm)
	Graphics (Inkscape)
Software	Gblocks, Geneious, MUSCLE, PartitionFinder, Phyluce, PRANK, SATé, SortaDate (multiple sequence alignment, partitioning, and filtering), ASTRAL, ExaBayes, IQ-TREE, MrBayes, PAUP*, RAxML, RAxML-NG, RevBayes (phylogenetic inference), BEAST (1 & 2), DPPDiv, Multidivtime, PAML, PhyloBayes, treePL (divergence time estimation), BAMM, PyRate (diversification rate estimation)
Methods	Bayesian inference, phylogenetics, morphological modeling, fossil tip-dating, posterior predictive simulations, species delimitation
Languages	Nepali (native)
	English (fluent)
	Hindi (fluent)
	German (basic)

References

Prof. Dr. Sebastian Höhna
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Dr. April M. Wright
Southeastern Louisiana University
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